The bimodal distribution of sizes may be real. Failure to observe this in the previous laboratory studies of returned material may be due to the limited observation of submicron particles. An infrared/ultraviolet photometric study of the lunar surface from the Zond 3 flyby spacecraft led to the conclusion that the roughness distribution was bimodal with peaks at 40 μm and at 0.6 μm (ref. 9).

For remote determinations of lunar soil grain sizes with Surveyor 3, several kinds of data were used. The photometric function of the surface was changed by imprint of a spacecraft footpad. With this information, Gault *et al.*[10] concluded that the intricate structure of the material was smaller than the resolvable 1 mm and larger than the 0.5 μm wavelength of visible light; they added, "in practice, the particles cannot be much larger than the order of 10 μm"[11]; this estimate was supplemented by Surveyor 3 photometric data; Christensen *et al.*[12] matched the photometric change with laboratory models of various grain sizes. They also noted that the lunar soil replicated ridges on the bottom of the footpad, about 50 to 70 μm high, and matched this with laboratory models. They deduced that a substantial percentage of the soil was smaller than 60 μm, and a significant fraction smaller than 10 μm. To Gold[13], the ridge pattern implied that less than 20% of the material was coarser than 50 μm, and the photometric change implied that the material was finer than 10 μm. Shoemaker and Morris[14] extrapolated downward the particle size distribution from above 1 mm. Depending on the assumptions made, they obtained a median size, by volume, of 15 to 88 μm, with quartiles of the distribution at 3 to 6 μm and at 126 to 850 μm. All but one of these determinations are in agreement with the laboratory measurements of soil from Surveyor 3 reported here; one[13] was low by 3 to 5 times[11].

Information acquired on the lunar surface from other Surveyor spacecraft was also used to determine soil particle size. Table 1 lists the results, as well as overall conclusions, from all of the Surveyor missions by various authors. For the most part, the findings from Surveyor are in good agreement with the results of the present work and with other measurements on lunar material returned to Earth.

The pertinent distribution curves given by authors cited in Table 1 are included in Fig. 3*a*. Differences in normalization account for much of the apparent difference from the present results at the larger sizes. The distributions derived from on-surface observations are significantly broader than the laboratory results, but are reasonable in view of the limited data on which they were based and the variation among returned Apollo samples.

It seems therefore that Surveyor 3 provided valid on-surface measurements of lunar soil particle size. Examination by television of areas disturbed by the spacecraft, when combined with laboratory modelling, gave the most reliable results; measurements of mechanical properties and certain extrapolations from sizes of larger, resolved particles, also gave good size estimates. This increases confidence that valid determinations of soil particle size can be made on the planets with Viking or other planetary landers.

We thank Kenneth Evans for electron microscopy, Michael Blackstone and John Addington for mathematical assistance, and Marylou Ingram for cooperation in experimental work. This work was sponsored by the National Aeronautics and Space Administration.

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Received December 18, 1972.

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Concordant Cretaceous Palaeolatitudes from Felsic Plutons in the Canadian Cordillera

Beck and Noson[1] report that three sites from a Cretaceous (≈85 m.y.) granitic batholith in the Cascade Mountains of central Washington, USA, give anomalously low palaeomagnetic inclinations compared to coeval rocks from the rest of the North American plate. They suggest that the discordance results from post-emplacement northward translation of the western Cordillera by about 25 arc deg relative to the North American craton. Here I report palaeomagnetic results from 17 sites of Cretaceous (≈95 m.y.) quartz diorites and granodiorites collected along Howe Sound in southwestern British Columbia. These plutonic rocks are geologically part of the Coast Crystalline Belt of the western Cordillera and topo-

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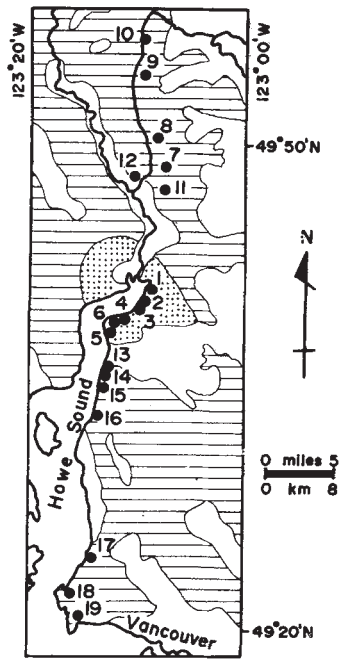


Fig. 1 General geology[8,9] of the Howe Sound area showing the "relatively old" and "relatively young" plutons by horizontal lines and stippling respectively. Sampling sites are shown as solid circles.

graphically part of the Coast Mountains which are the northward continuation of the Cascade Mountains. The pole position for these sites is not significantly different from Cretaceous positions for units from cratonic North America and therefore does not support the hypothesis of northward tectonic translation of the western Cordillera relative to the craton.

Samples were collected from two types of felsic plutons (Fig. 1). The contacts of the "relatively old" plutons[2] (sites 7–19) conform roughly to the NW–SE trend of the Coast Crystalline Belt. They are composed of quartz diorite and granodiorites containing variable amounts of biotite and hornblende[3]. Commonly they have abundant inclusions, are gneissic or banded, and are broken by a complex pattern of closely-spaced joints. In the Howe Sound area these "older" plutons both intrude and are overlain by metasediments containing Early Cretaceous fossils and by Upper Cretaceous sediments (Senonian, ≈80–85 m.y.)[3–6]. The "older" plutons were seemingly formed during the Cretaceous recrystallization and metasomatism of pre-existing rocks in an abyssal environment on a massive scale[3]. The available K–Ar radiometric ages on biotite of 93, 95, 95, 99 and 158 m.y. and on hornblende of 97 m.y. (refs. 2, 7) in the Howe Sound area are equivocal because they may reflect either the time of intrusion or of metamorphic up-dating. In either case, the characteristic remanent magnetization (RM_c) (the remanence isolated after partial alternating field (AF) demagnetization with a consistent direction) of these rocks was probably formed about 95 m.y. ago. The Squamish granodiorite (sites 1–6) is a "relatively young" pluton exhibiting discordant contacts[8]. It is an equigranular, unaltered, massive rock with rare inclusions and a simple pattern of widely spaced joints[2]. These "younger" plutons were apparently formed by the permissive intrusion of magmas to a high crustal level during the Upper Cretaceous. Three K–Ar radiometric age determinations on biotite give a mean of 92±4 m.y. and are consistent with this petrogenetic hypothesis[2]. If the "younger" pluton rocks retain an RM_c, they probably acquired it at that time.

At each of the 19 sites, 4 or 5 rock cores were drilled and oriented by Sun compass. The natural remanent magnetization (NRM) was measured[10] for 2 specimens from each of the 92 cores. One pilot specimen per site with an average NRM was demagnetized in an AF (ref. 11) in 8 steps up to 800 oe. Guided by its remanence intensity decay curve and stability index[12], I used an optimum AF intensity of 100, 200 or 300 oe to clean the rest of the specimens from its site before measuring their RM_c. The mean RM_c directions from 21 cores were rejected; 4 had too weak RM_c intensities, 14 had divergent (>20°) specimen RM_c directions, 2 had anomalous mean RM_c directions, and 2 were the sole cores remaining and hence insufficient to be representative of their site. This left 17 reliable mean RM_c directions utilizing data from 71 cores. These 17 sites have precision parameter estimates (k, ref. 13) between 21 and 540 with an average of 118, and cones of 95% confidence ($P_{0.05}$) with radii between 2.7° and 16.3° with an average of 8.2°.

The angular variance ratio method for analysing RM_c directions compares the between-group to within-group angular variance ratio (δ_b^2/δ_w^2) to the corresponding 95% probability statistic ($F_{0.05}$). The 3 geographic groups of "older" pluton sites [(7, 9–12) (13, 14, 16) (17–19)] are not statistically distinct, $\delta_b^2/\delta_w^2 \ll F_{0.05}$. This result suggests that the sites acquired their RM_c at essentially the same time during granitization and that subsequent between-group tectonic rotations have not been significant. Similarly the mean RM_c directions for the "older" pluton sites are statistically indistinguishable from the "younger" Squamish granodiorite sites' (1–6) directions, $\delta_b^2/\delta_w^2 = 1.06 \ll F_{0.05} = 3.32$. This indicates that both "older" and "younger" plutons acquired their RM_c at the same time—presumably during emplacement 90 to 100 m.y. ago—and that all 17 site mean RM_c directions may be combined to calculate the Howe Sound plutonic mean direction (declination 350.8°; inclination 65.2°; length resultant vector 16.1273; $k = 18.3$; $P_{0.05} = 7.9°$) and the resulting pole position of 129.2° E, 83.4° N ($\delta_m = 12.9°$, $\delta_p = 10.4°$).

I used the angular variance ratio method to compare the Howe Sound mean pole position to the reliable Cretaceous pole for North American units (Table 1, poles 1–7, Fig. 2). In determining the mean pole position for the Cretaceous for North America, I have excluded the positions determined for the Franciscan basalts[15] and ultramafics[16] because they may be Jurassic and have probably been tectonically rotated. Similarly the positions for the Mt Ascutney ring dike[17], the Stevens Pass granites[1], the Mesaverde sediments[18], and the

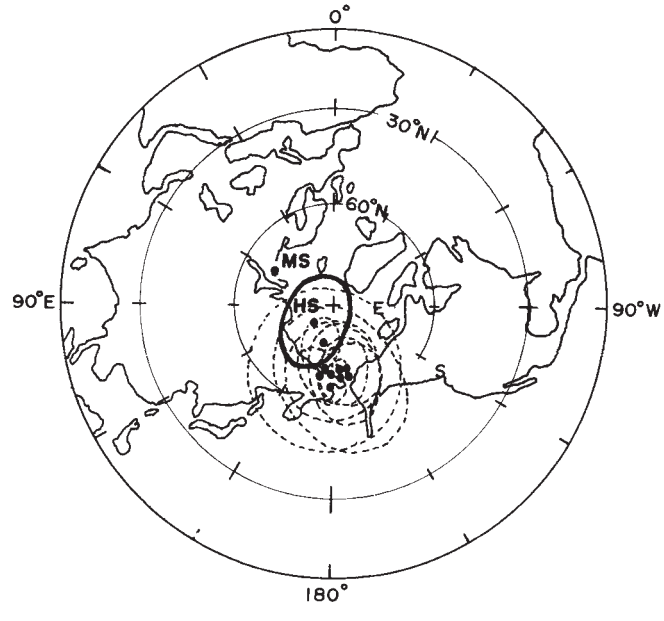


Fig. 2 Palaeomagnetic pole positions from North American Cretaceous rock units. The pole positions are shown as solid circles with their circumscribing oval of 95% confidence as listed in Table 1. HS is the Howe Sound pole position, E is the present magnetic pole position, and S is the sampling site location.

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Table 1 Palaeomagnetic Pole Positions from North American Rock Units of Cretaceous Age

Rock unit	Locality	Age (m.y.)	Sites	Pole position Long.	Lat.	δ_p	δ_m	Reference
Monteregian Hills	SE Quebec	124±3	32	171	71	2	3	20, 21
Mt Megantic intrusion	SE Quebec	113	6	189	68	7	10	21, 22
Isachsen diabase	Canadian Arctic	106±5	10	180	69	14	14	23
Alkalic complexes	Arkansas	97±10	19	170	70	9	13	24
Sierra Nevada pluton	California	86±4	14	165	69	10	10	25
Elkhorn Mt volcanics	SW Montana	80	13	171	69	7	10	26
Boulder batholith	Montana	75±7	15	193	76	6	7	26
Howe Sound plutons	SW British Columbia	94±6	17	231	83	10	13	This study

Long., Lat., denote longitude west and latitude north in degrees.
δ_p, δ_m are the semi-axes of oval of 95% confidence about the pole along and perpendicular to the site-pole great circle in degrees.

Dakota sandstone[19] were excluded because of insufficient sampling and/or large error limits. The computed Cretaceous pole position is 176.4° W, 70.5° N (error $P_{0.05}=2.9°$). The resulting variance ratio of 3.32 is only slightly greater than the appropriate $F_{0.05}$ (3.21) and much less than the 99% confidence statistic ($F_{0.01}$) of 5.26. Unfortunately the test is being forced by comparing site mean poles to rock unit mean poles, by comparing two different population distributions ($P_{0.05}$ of 10.0° against 2.9°), and by accepting the low error estimate of 2.9° for the Cretaceous which contrasts with the higher root mean square estimate of 10° using the individual Cretaceous pole position errors. The test is inconclusive and the Howe Sound pole position is not statistically anomalous compared to other reliable Cretaceous pole positions.

There are several possible explanations for the poor agreement between the Cretaceous mean and Howe Sound pole positions. First, the latter may retain Cainozoic remanence components although the exclusion of the present magnetic pole from its cone of 95% confidence militates against this alternative. Second, the Howe Sound diorites may retain some Jurassic remanence components but the complete reorganization of the plutonic rocks makes this unlikely. Third, the Cretaceous magnetic pole might have a somewhat greater apparent wander relative to North America than is suggested by the tight clustering of known pole positions. Fourth, a 35° clockwise rotation (about an adjacent nearly vertical axis) of the Howe Sound area plus a 9° northward translation of the Cordillera relative to cratonic North America would produce agreement. But no structural or other trend direction changes are evident to support such a rotation. Finally, a 15° SW tilt of the Howe Sound area about the NW–SE trending horizontal axis parallel to the Coast Crystalline Belt seems the most plausible alternative. Unfortunately contemporaneous or younger sediments overlying the plutons have been eroded so that the hypothesis is difficult to test. The Howe Sound area is on the southwestern side of the Coast Range which has been a focus of Cainozoic uplift. Also, the Range has been subjected to some Cainozoic SW–NE compression resulting from the collision of the North American and Farallon plates[32].

There are several reasons to question the Beck and Noson[1] hypothesis of northward translation of the western Cordillera by 25 arc deg. First, their data base of 3 sites is probably insufficient as shown by more extensive palaeomagnetic studies of acidic intrusive rocks[27,28]. Second, younger Eocene[29], contemporaneous Cretaceous[25] and older Jurassic[28] rock units from within the Insular and Coast Crystalline Belts of the Cordillera give clearly concordant pole positions indicating no relative rotation or translation between the Cordillera and craton. Third, Monger and Ross[30] give three models to explain fusulinacean distribution in Cordilleran Palaeozoic rocks of which the simplest model invokes palaeoenvironmental factors rather than tectonic translations. Fourth, if a large translation has occurred, overwhelming geological and geochemical evidence suggests that it was pre-Mesozoic[31].

I conclude that the palaeomagnetic results from the Howe Sound plutons do not support the concept of some 25° of northward translation of the western Cordillera relative to cratonic North America but may indicate minor tectonic deformation during the past 100 m.y.

I thank R. L. Wright who assisted in the sample collecting and specimen preparation, G. Freda who made the remanence measurements, and Dr W. F. Fahrig and Dr J. H. Foster who read the manuscript.

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Received October 24, 1972.

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